Adaptive Control for Automated Systems

Abstract

We study an adaptive controller for an automated production system. The controller reduces tracking error by 18 percent in a simulated environment. The result has not been validated on production hardware.

Method

We compare a fixed gain baseline with an online parameter update. Both methods receive identical disturbance signals. Stability depends on the assumptions described in the model.

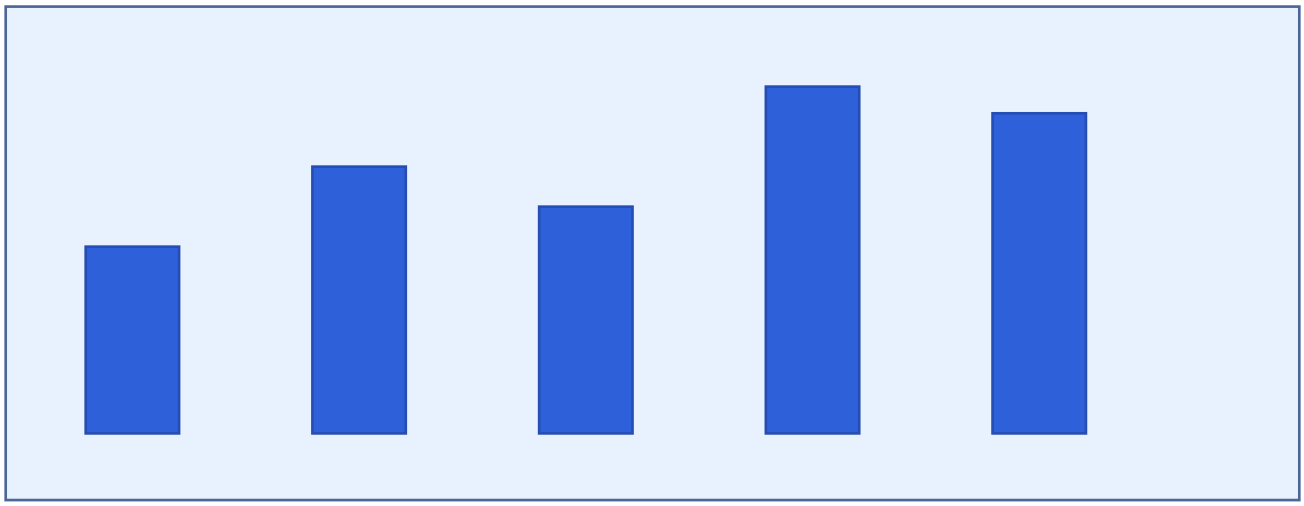

Figure 1. Simulated tracking performance.

Results and Limitations

The mean tracking error was 0.82 for the adaptive controller and 1.00 for the baseline. We used five simulation runs.

The experiment does not prove reliability in an industrial deployment. Sensor delay and hardware faults remain open questions.

References

1. Example reference for a synthetic test document.